



Sales Forecasting and Demand Prediction – EDA

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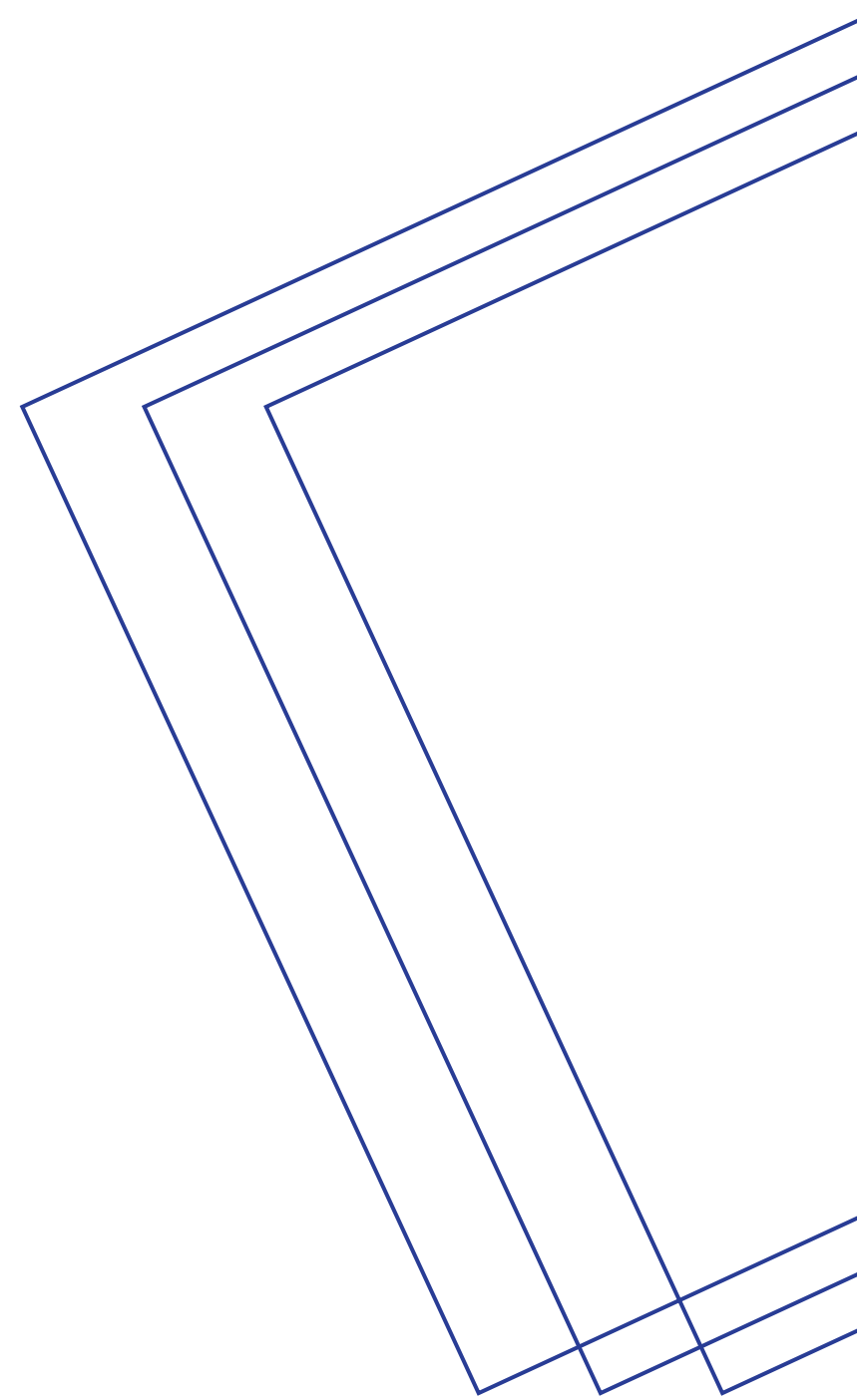
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01 Introduction

Sales Forecasting Analysis

This project aims to analyze sales trends and predict future sales using historical data to support data-driven decision-making.

Key Questions

- What are the key factors affecting sales trends?
- How can we improve sales forecasting accuracy?
- What actionable insights can help optimize business strategies?

Methodology

- Data cleaning and preprocessing.
- Exploratory Data Analysis (EDA).
- Deriving actionable insights.

02 Dataset Overview

Data Source

This dataset is real data of 9,800 records collected from a private Super Store provider. [Dataset Link](#)

Challenges

- Imbalanced distributions in Target Sales.
- Outliers Termination

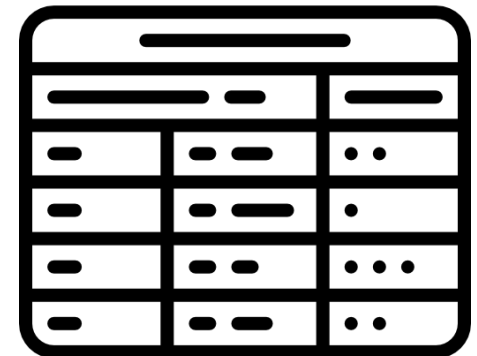
Key Features

- **Sales:** Target variable.
- **Time-Based Features:** Order Date & Ship Date.
- **Product Name:** Identifier for each product.
- **Category:** Product Category.
- **Region, State & City:** Geographical Features.

03 Data Cleaning and Preprocessing

Columns Manipulation

Removed The identifier columns that were irrelevant to the Analysis Row ID, Customer ID & Product ID

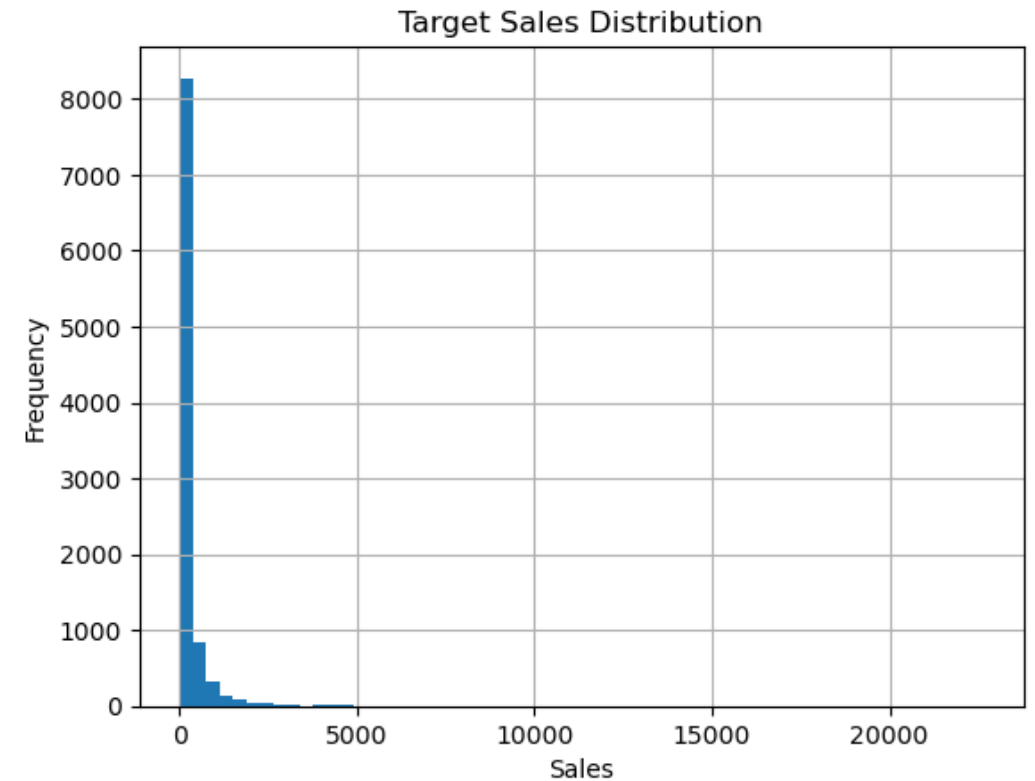


03 Data Cleaning and Preprocessing

Outliers

The Analysis Showed an Imbalance in the Distribution of Target Variable “Sales”

Which was Eliminated Using Feature Scaling (Normalization & Standardization)



03 Data Cleaning and Preprocessing

Feature Engineering

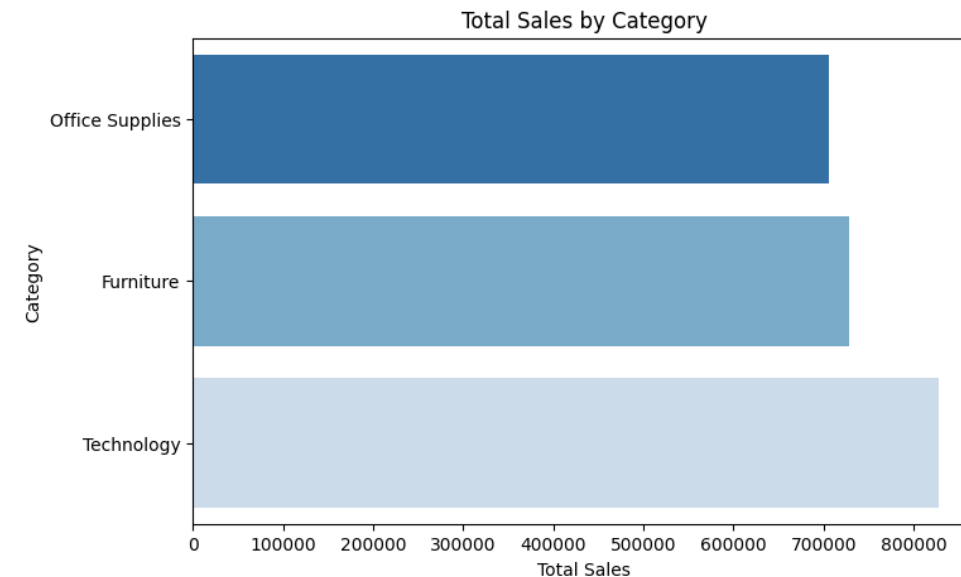
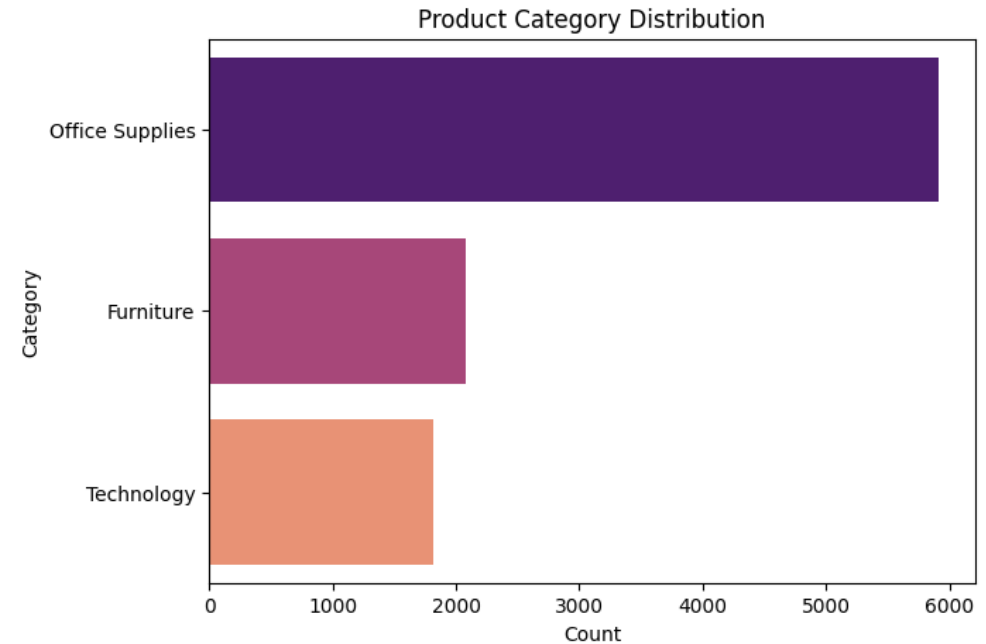
- **Encoded Categorical Variables (One-Hot & Label Encoding):**
Ship Mode, Segment ...etc.
- **Maintained a Unified Date Format across features “%d/%m/%Y”.**
And extracted Temporal features from the date column (Day, Month, Year)
- **Obtained an Aggregated Sales Performance per Product Feature**

04 EDA Insights

Product Analysis

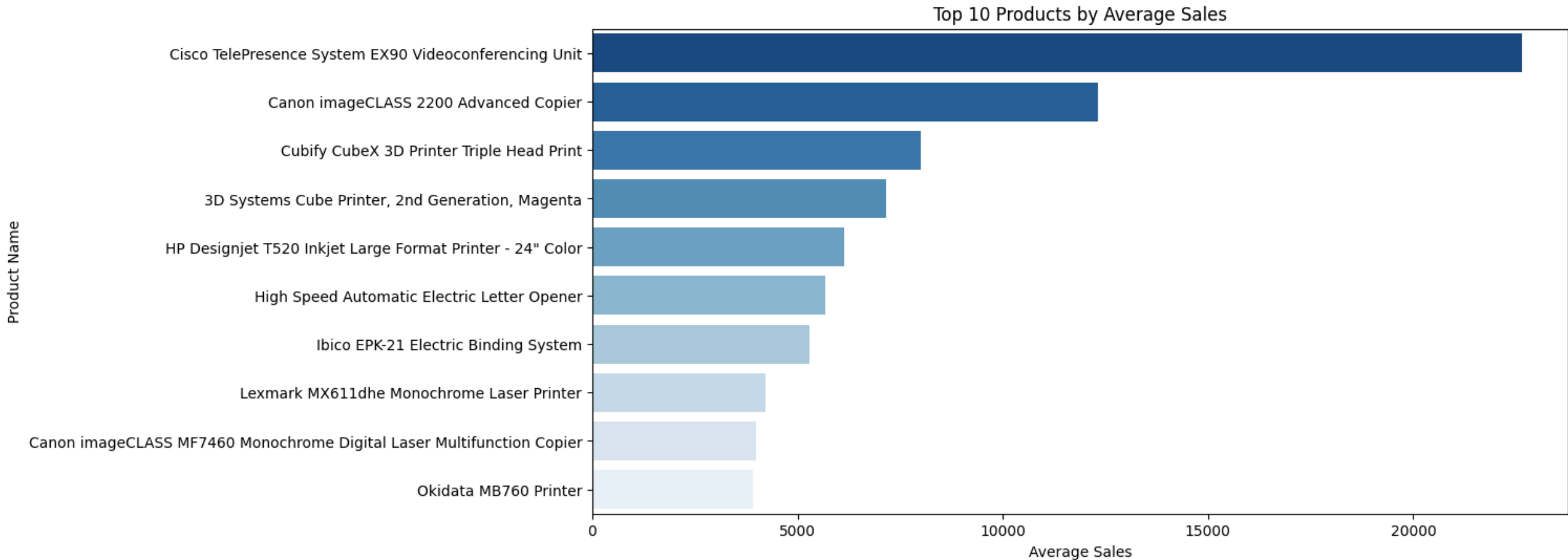
The analysis indicates that the **Technology product category** generates the highest sales, despite the majority of offered products belonging to the **Office Supplies category**.

This suggests that **expanding the range of technology-related products** could lead to **higher future sales and revenue growth**.



04 EDA Insights

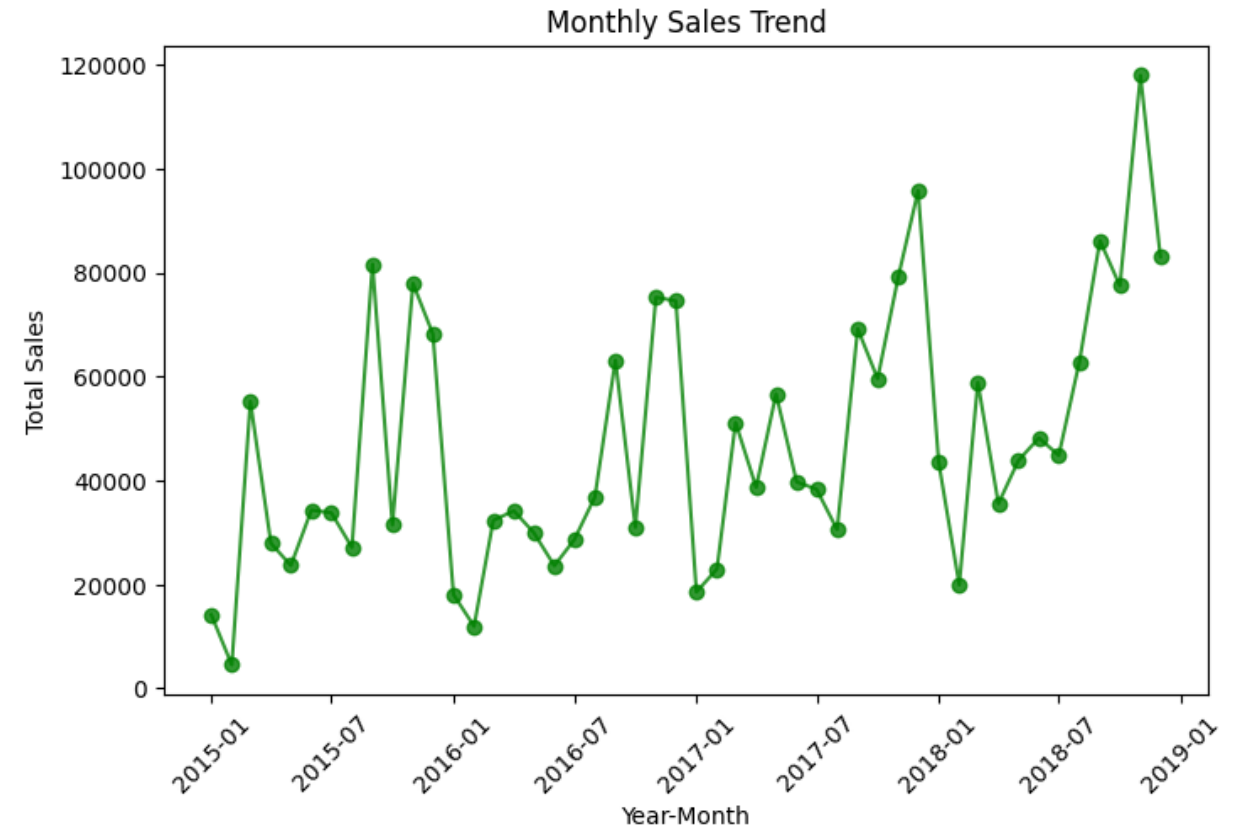
Top 10 Products Sales



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Time Series Analysis

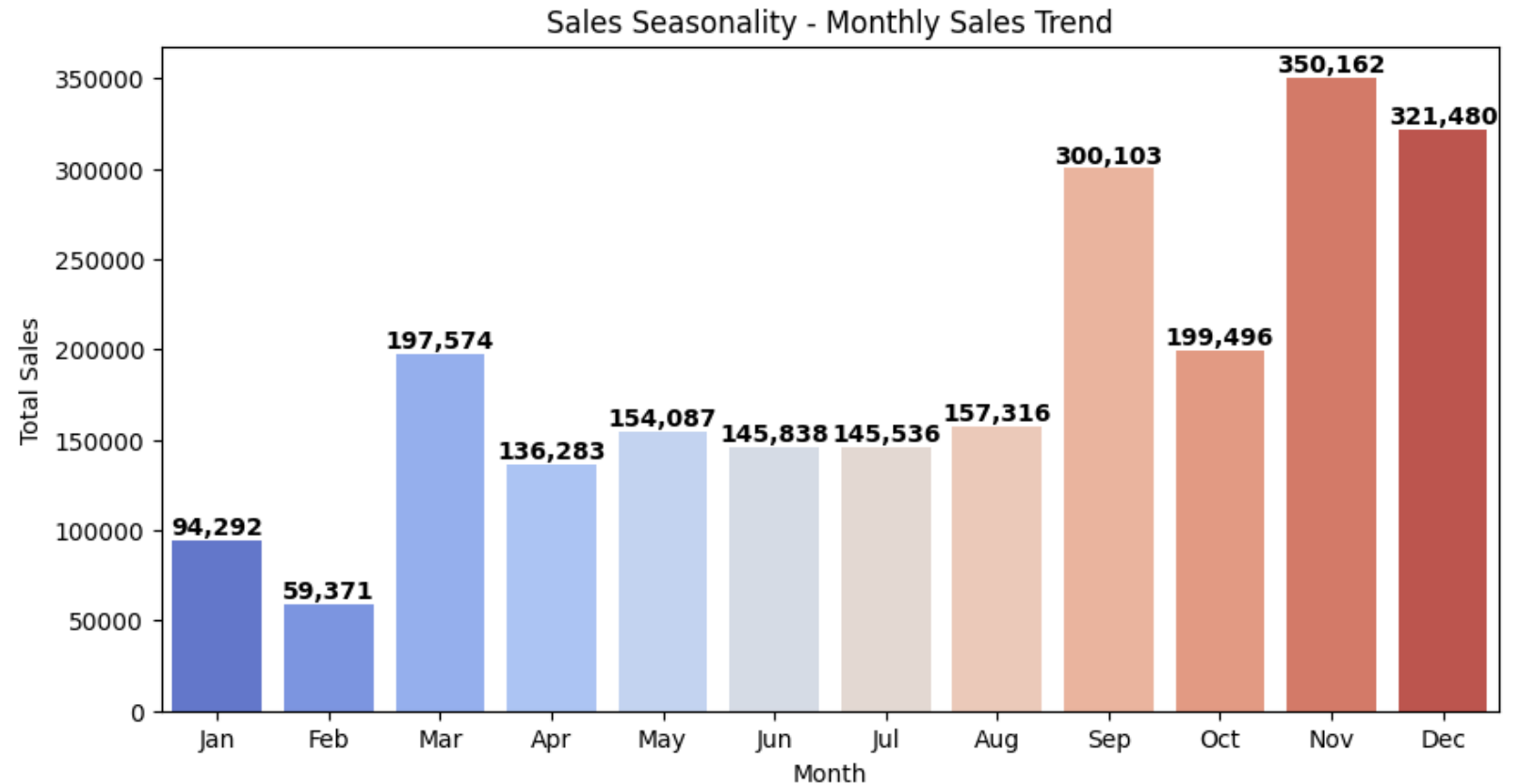
Peak sales occurred
between **2018** and
2019.



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Time Series Analysis

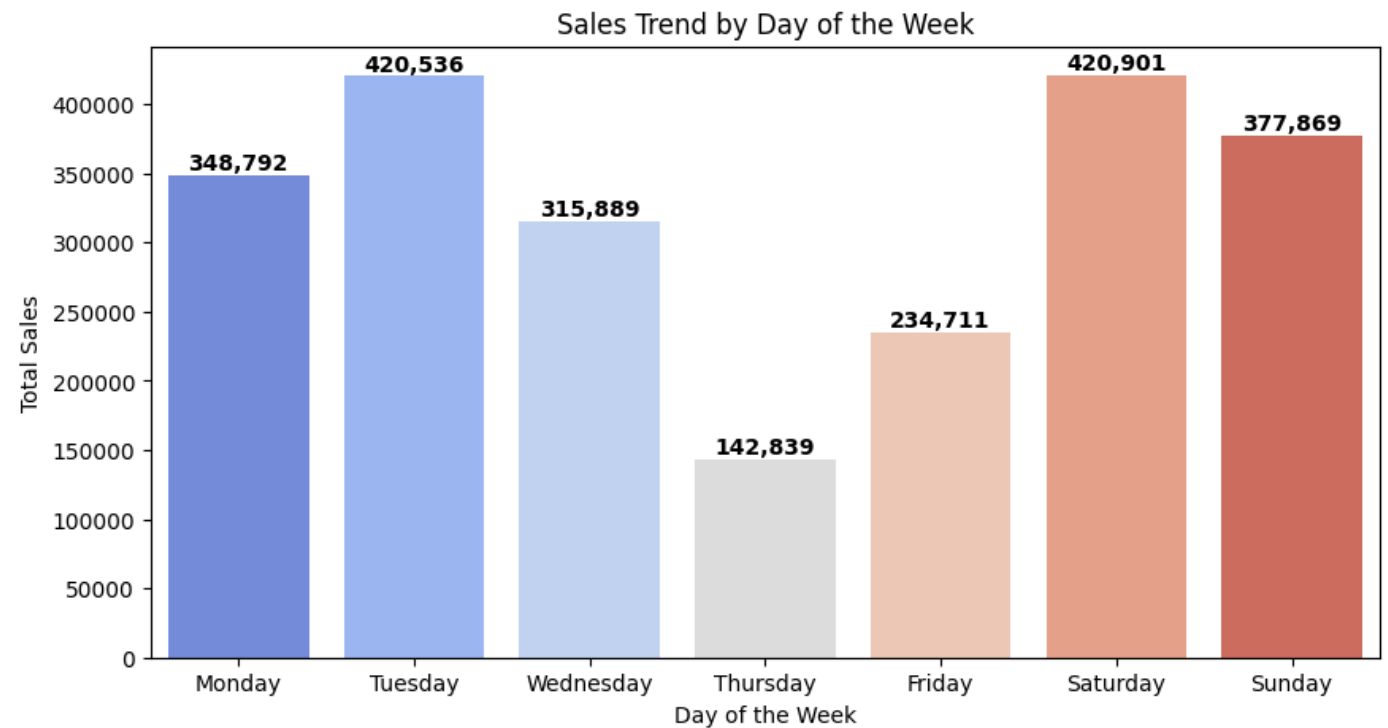
Peak sales occur in November, suggesting the need for targeted marketing and optimized inventory to maximize revenue.



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Time Series Analysis

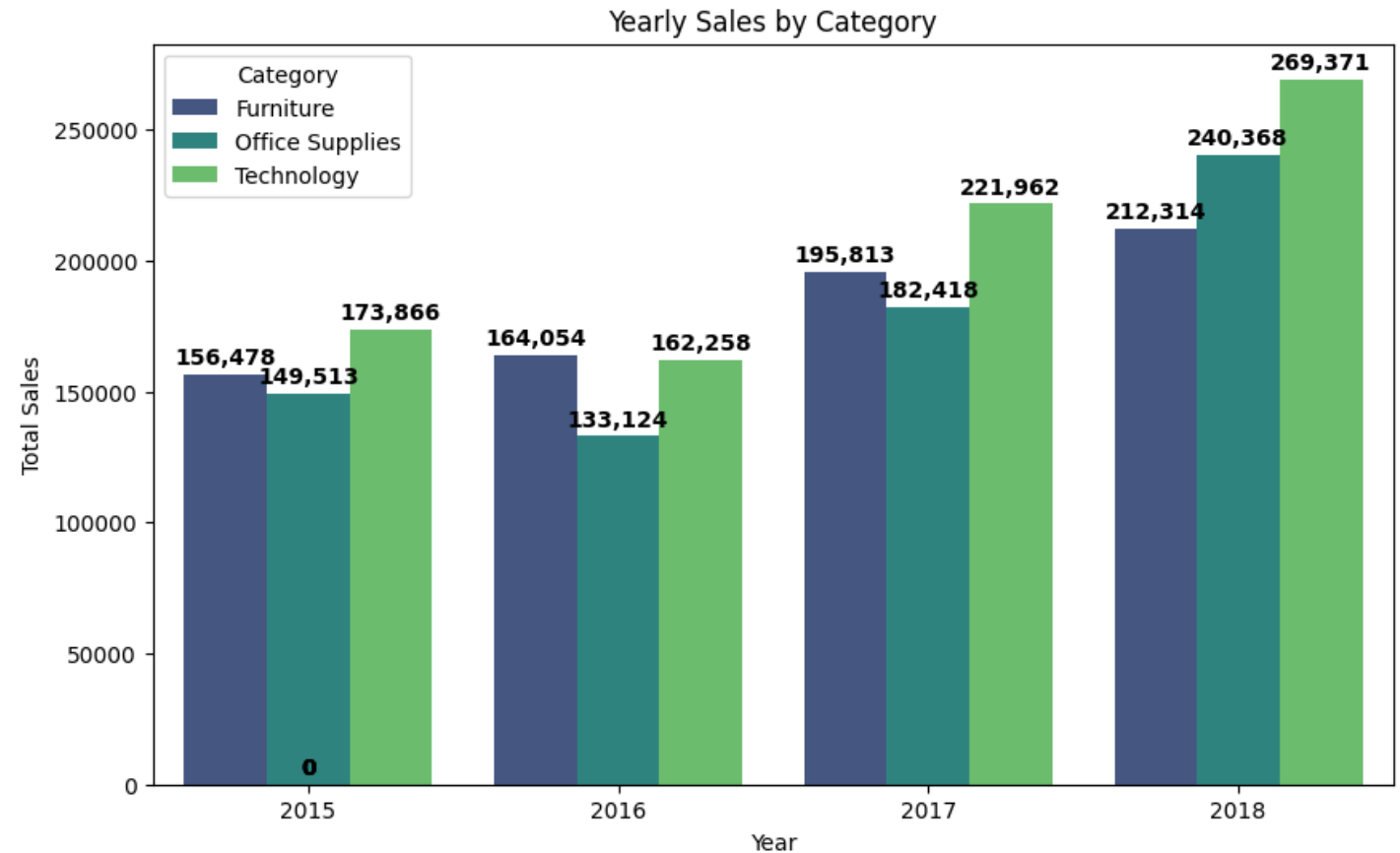
Highest sales occur on Saturdays, suggesting a strategic focus on weekend promotions and inventory management to maximize revenue.



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Time Series Analysis

Highest Sales For Tech Category, This suggests that expanding the range of technology-related products could lead to higher future sales and revenue growth.



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Geographical Analysis

Top 3 Cities in sales are

25.1% in
New York

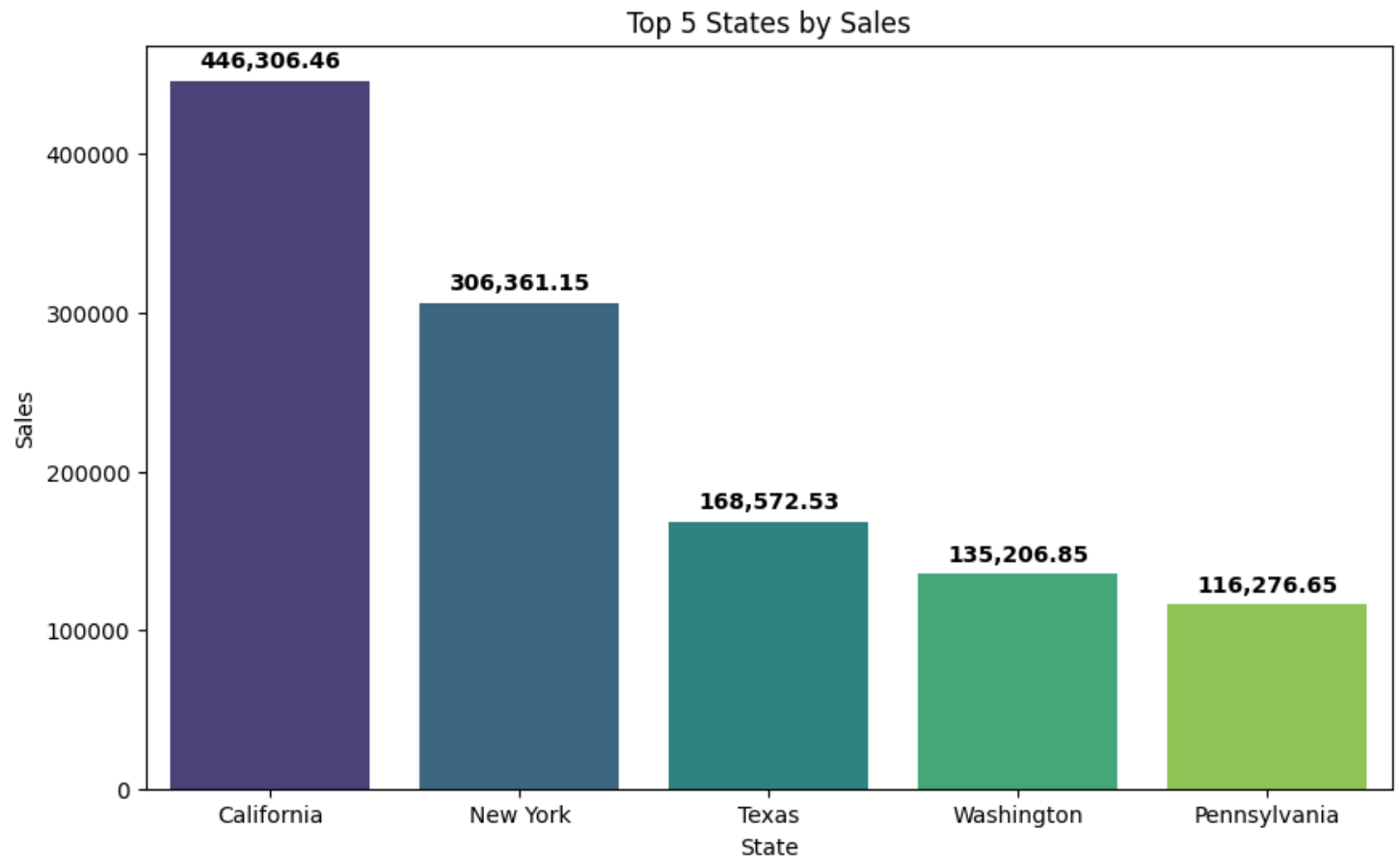
17.2% in
Los Angeles

11.5% in
Seattle

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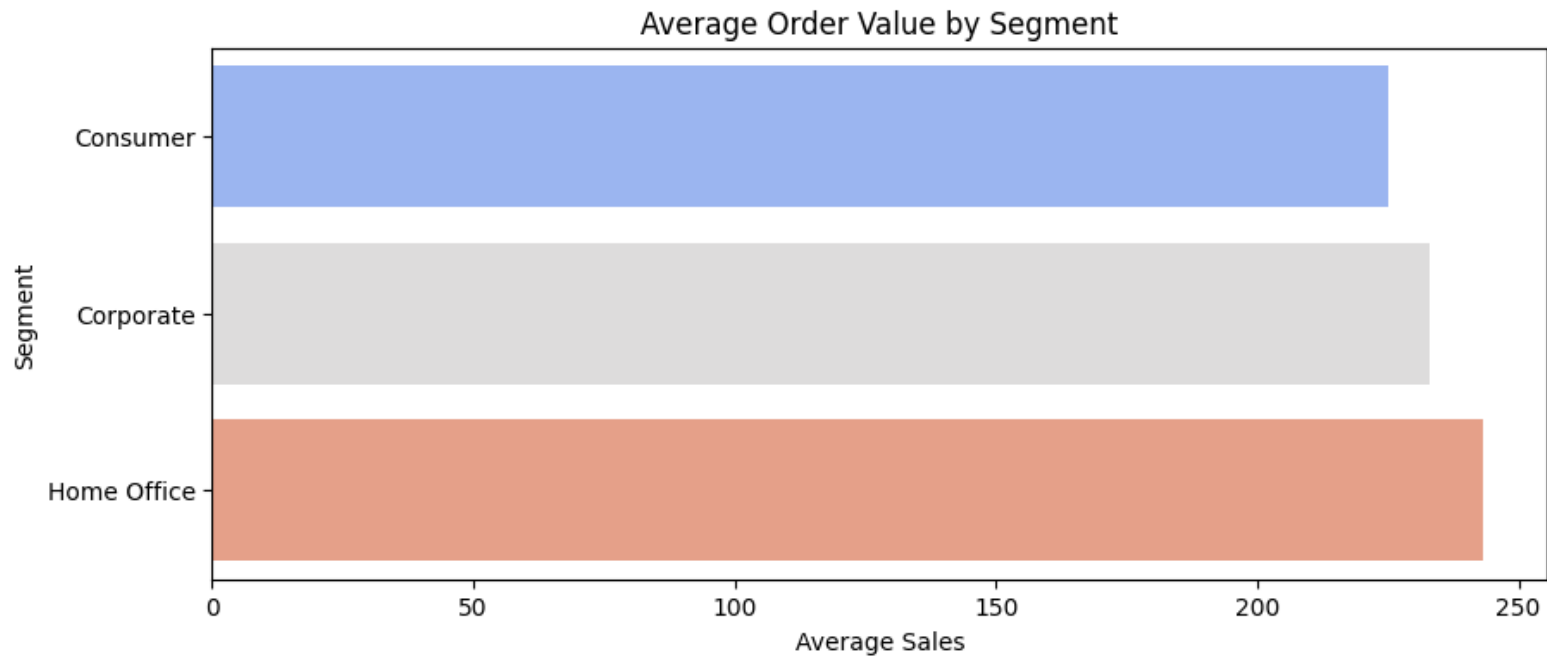
Geographical Analysis

California records the highest sales, indicating a strong market presence and potential for further expansion.



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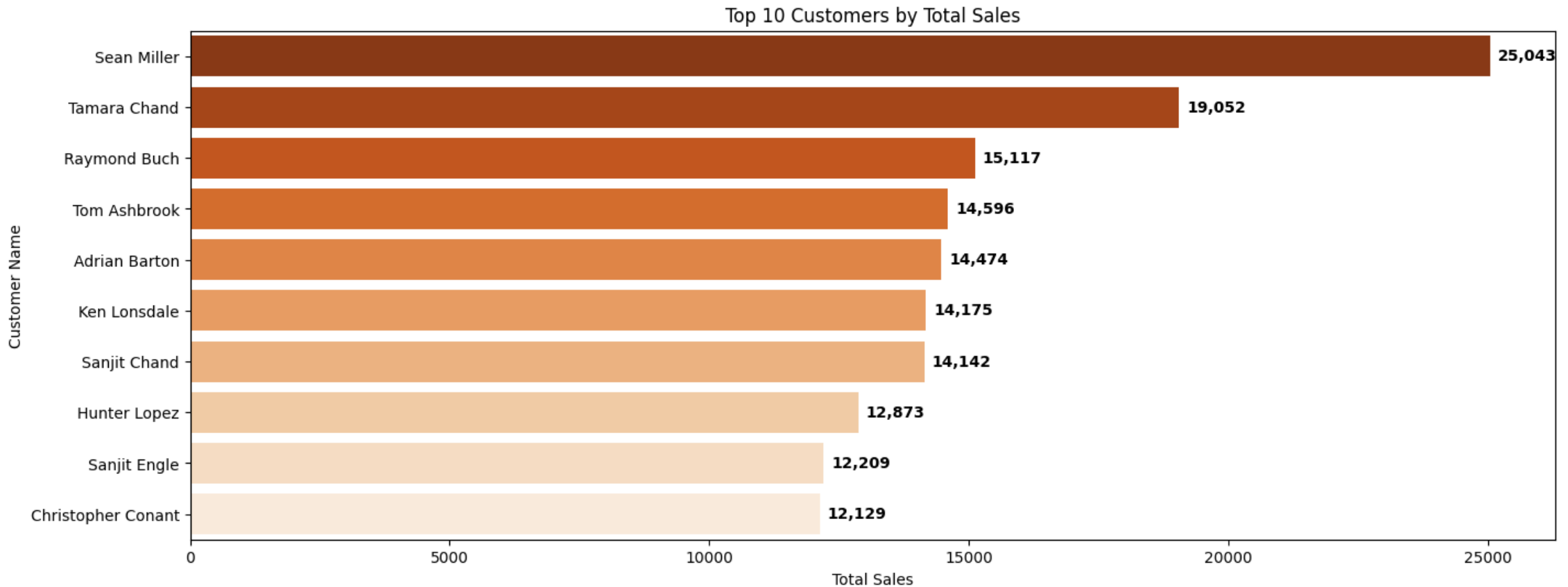
Customer Analysis



The Home Office segment generates the highest sales, suggesting a key target audience for future strategies

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Customer Analysis



Questions

Thank You

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